A Framework for Detecting Hazardous Events

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ABSTRACT

Using video analysis for detecting hazardous events such as fire/smoke activity, impending threats, or suspicious behaviors has spurred new research for security concerns. To make such detection reliable, researchers must overcome difficulties such as classification by the importance of consequences, imbalances of positive and negative data, environmental factors, and variation in camera capabilities. This paper puts forward a general framework for hazardous event detection which includes spatial-temporal feature extraction, statistical-based classification for biased data and calibration for environmental change. At the current stage of development, the framework can work effectively for detecting hazardous events like fire/smoke from video sequences.

Keywords: video analysis, hazardous events detection, feature extraction

1. INTRODUCTION

The heightening of concerns about national security has intensified the need for effective, efficient surveillance. Traditionally, sensors have been used for detecting events. Though useful for simple event detection such as temperature changes, presence of carbon dioxide, and motion, sensors often suffer from low specificity. For instance, a carbon-dioxide-based fire sensor cannot differentiate the CO_2 emitted from real fires from that of faulty gas appliances or vehicle exhausts. Moreover, tasks such as detecting intruders and suspicious vehicles require sophisticated cues that sensors cannot collect.

With the proliferation of inexpensive cameras and the deployment of high-speed, broad-band network, it has become economically and technically feasible to employ cameras for event detection. Mapping visual cues to events presents many research challenges. In this paper, we limit our scope to fire and smoke events. We focus on presenting a framework that can achieve effective feature-to-event mapping. Specifically, our feature-to-event mapping framework aims to tackle two research problems: *biased statistical inference* and *inconsistent visual features*.

1. *Biased statistical inference.*

   We propose learning methods (in Section 2.2) to deal with the following two inference constraints:

   - *Classes of unequal importance.* For hazardous event detection, misclassifying a positive event (false negative) incurs more severe consequences than misclassifying a negative one (false positive).
   - *Imbalance in training data.* Positive events (hazardous events) are always outnumbered by negative events in the training data. In an imbalanced set of training data, the class boundary tends to skew towards the minority class and becomes very sensitive to noise. Hence the rate of false negatives increases.

2. *Inconsistent features.*

   We discuss calibration strategies (in Section 2.3) for tackling the inconsistent feature problem caused primarily by the following two sources:

   - *Different camera capabilities.*
     Different cameras generate videos of different color intensity and texture quality. Furthermore, camera movements (e.g., panning and zooming) can affect video features, as can the distance between cameras and the areas of surveillance.

- *Variable environment factors.*
  External environment factors such as lighting conditions (e.g., day or night) and weather (e.g., foggy or raining) also affect video features.

The rest of the paper is organized as follows: Section 2 introduces a framework for hazardous event detection and describes its key steps. Section 3 compares the experimental results of different statistical learning methods for detecting fire and smoke events. Section 4 presents some related work in video analysis and event detection. Finally, we provide concluding remarks and put forward ideas for future work in Section 5.

2. EVENT DETECTION FRAMEWORK

To tackle the challenges indicated in Section 1, Figure 1 presents our framework for event detection. The framework consists of a training phase and a detecting phase, and they together are composed of six major steps.

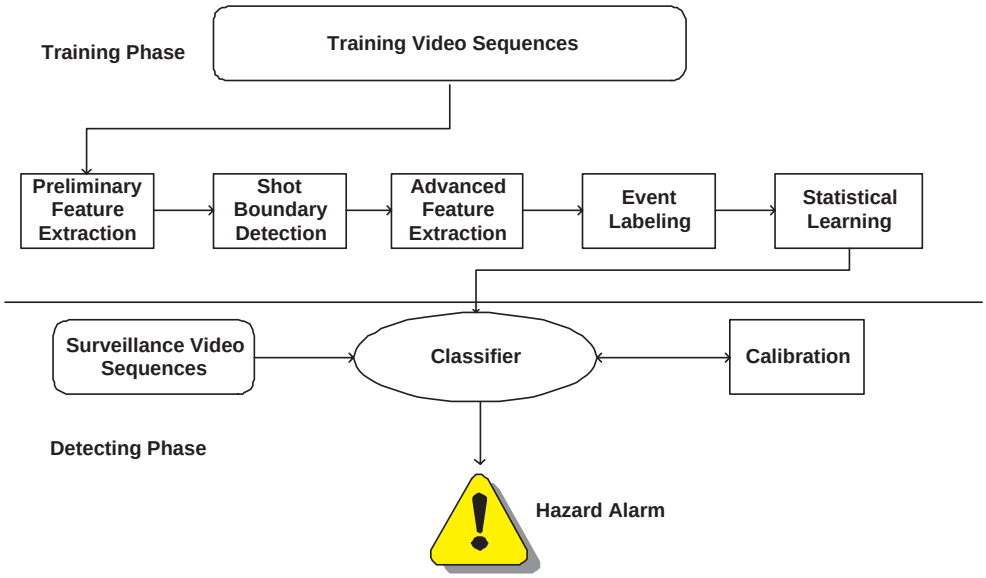


Figure 1. Framework.

1. *Preliminary feature extraction.* This step extracts intra-frame low-level features including color, texture, and shape information.

2. *Shot boundary detection.* Based on the feature difference between frames, this step performs *shot boundary detection.* (We use the algorithm proposed in[1] for identifying shot boundaries.) At the end of this step, a video sequence is divided into shots.

3. *Advanced feature extraction.* Once shots are identified, we extract inter-frame features for each shot. The additional features extracted in this step are temporal variations between frames, which characterize motion patterns of a shot.

4. *Event labeling.* We use the shot as the basic training unit and label each as positive (hazardous) or negative (benign).

5. *Statistical learning.* We employ statistical learning algorithms to train a binary classifier for detecting hazardous events. The trained classifier processes surveillance video sequences and generates alarms when hazardous events occur.

6. *Calibration.* Finally, a calibrator adjusts class prediction when it finds changes in camera capabilities or environmental factors.

In the remainder of this section, we focus our discussion on three steps: *feature extraction*, *statistical learning*, and *calibration*.

2.1. Feature Extraction

Feature extraction for event detection must fulfill two design goals: *adequate representation* and *efficient computation*. Adequate feature representation is the basis for modeling events accurately. Low computational complexity is equally critical, since multiple frames from multiple cameras may need to be processed simultaneously.

To achieve a reasonable balance between effectiveness and efficiency, we extract the following intra-frame and inter-frame features (Table 1 summarizes the features):

1. *Intra-frame features.*

   For each frame, we extract color, texture, and edges. We first divide color into twelve color bins, of which eleven bins are for the eleven culture colors[2] and one remaining bin for outlier colors. For each culture color, we extract color features of multiple resolutions, which include color histograms, color means and variances in HSV channels, color spreadness, and color-blob elongation. For texture extraction, we use discrete wavelet transformation because of its computational efficiency. Texture descriptors such as texture elongation, texture spreadness, energy means and energy variances are calculated in three orientations (vertical, horizontal, and diagonal) and three resolutions (coarse, medium, and fine). Finally, we extract edges using the Canny method.[3] The per-frame feature extraction time on a Pentium III 1.7G HZ processor is 100 to 200 milliseconds. This extraction method is much faster than some other algorithms discussed in Section 4. For example, Gabor texture extraction, typically takes a few seconds per frame.

2. *Inter-frame features.*

   Inter-frame features characterize object motion and can be helpful for discerning object types. For example, flame and sunlight have different motion patterns, although they may have similar low-level color and texture features.[4] Without loss of generality, we define a hazardous event as a temporal process, wherein temporal variations of intra-frame features are used to describe motion patterns. Specifically for fire and smoke, we model their motion patterns by the χ^2 distribution of inter-frame difference as follows:

   Let F_i be the ith frame in the video frame sequence. Each F_i is represented by a D-dimensional feature vector $\{f_{i,1}, ...f_{i,j}, ..., f_{i,D}\}$, in which $f_{i,j}$ denotes the jth feature of frame F_i. Given two frame F_i and F_k, we define the feature distance on j^{th} feature between these two frames as

   $$d_j(i,k) = |f_{i,j} - f_{k,j}|. \tag{1}$$

   We use Gaussian normalization to convert $d_j(i,k)$ into the range of [0,1]. The sum of $d_i(i,k)^2$ follows the χ^2 distribution. Two events are considered similar if they have a similar distribution in the temporal scale (denoted as t_d). The temporal scale represents the sampling frame interval. For fire and smoke, since their movement is not very radical, we set $t_d = 5$. Medioni[5] has conducted experiments to prove that different events exhibit different motion patterns. Typical motion patterns of fire and smoke are depicted in Figure 2, in which the x-axis represents frame differences calculated at the specified temporal scale-t_d and the y-axis indicates the frequency.

2.2. Statistical Learning

Support Vector Machines (SVMs) have been successfully used in many applications such as object recognition[6] and handwriting recognition.[7] In this paper, we use SVMs as our classifier to identify hazardous events. We first provide a brief overview of SVMs to set up sufficient context for discussion. We then describe the causes of misdetecting rare events. Finally, we propose methods that improve SVMs for detecting rare events.

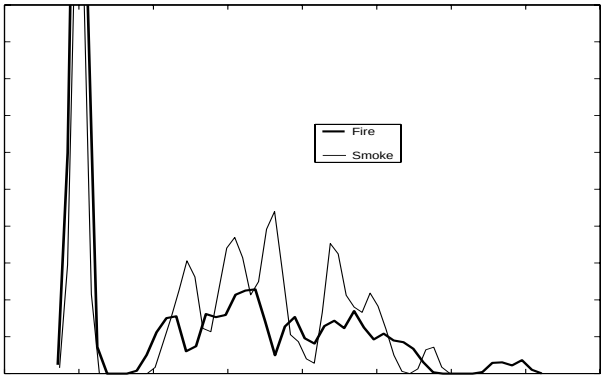


Figure 2. Fire/smoke Temporal Pattern.

Feature Type	*Feature Name*
Color	*Histogram*
	Mean
	Variance
	Spread
	Elongation
Texture	*Texture elongation*
	Texture spreadness
	Energy mean
	Energy variance
Edge	*Canny edge*
Temporal variance	$Inter - frame\ difference\ distribution$

Table 1. Feature Summarization.

2.2.1. SVM Overview

We consider SVMs in a binary classification setting. We are given a set of training data with labels $\{(x_1, y_1), (x_2, y_2), ...(x_n, y_n)\}$, where x_i is the ith training instance and y_i its label, either -1 for benign events or 1 for hazardous events. SVMs separate these two classes by a hyperplane with maximum margin.[8] As Figure 3 depicts, the separating hyperplane is located between the two classes. The training instances that are located closest to the hyperplane are *support vectors*.

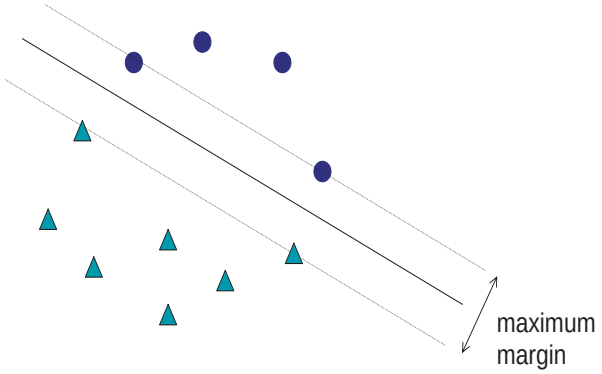


Figure 3. Simple Support Vector Machines.

For nonlinearly separable cases, SVMs can project the training data onto a higher dimensional feature space via a Mercer kernel operator K. In addition, Vapnik's soft-margin theory[9] introduces slack variables ξ_i to permit training error. Given x_i, SVMs model class prediction as

$$y_i(w \cdot \Phi(x_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, i = 1, ..., n, \tag{2}$$

where w is the norm to the hyperplane, $|b|/||w||$ is the perpendicular distance from the hyperplane to the origin, and Φ is an input-to-feature space mapping function. The optimal solution of SVMs is formulated by maximizing the Lagrangian

$$L_D = \sum_{i=1}^{p} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{p} \alpha_i \alpha_j y_i y_j K(x_i, x_j), \tag{3}$$

subject to the following constraints:

$$0 \leq \alpha_i \leq C \quad and \quad \sum_{i=1}^{p} \alpha_i y_i = 0, \tag{4}$$

where C is a constant larger than zero used as penalty for soft-margin SVMs. According to the KKT conditions, the value of α_i has three ranges:

$$\alpha_i = 0, \qquad Non - support\ vectors, \tag{5}$$

$$0 < \alpha_i < C, \qquad Support\ vectors\ and\ \xi_i = 0,\ and \tag{6}$$

$$\alpha_i = C, \qquad Support\ vectors\ and\ \xi_i > 0. \tag{7}$$

By solving Equations 3 and 4, the class prediction function is formulated as

$$f(x) = sign(\sum_{i=1}^{p} \alpha_i y_i K(x, x_i) + b). \tag{8}$$

2.2.2. Misclassification Causes

As discussed in Section 1, hazardous event detection challenges SVMs at two fronts: unequal class importance, and imbalance in training classes. To alleviate the first problem, a simple method is to shift the class boundary towards the benign-class side so that the false negative rate can be reduced. However, this naive method tends to reduce false negatives at the expense of increased false positives. Figure 4 shows the effect of moving the boundary. A desirable solution (presented in the next section) is to reduce false negatives while keeping false positives low.

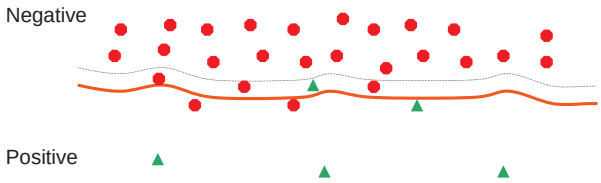


Figure 4. Change Threshold.

A subtle but severe problem that a SVM classifier faces is the skewed class boundary caused by imbalanced training data. As the number of negative examples (the majority class) grows, so does the number of negative support vectors that exert influence on an unlabeled instance. To illustrate this problem, Figure 5 depicts a 2D checkerboard example. The checkerboard divides a 200*200 square into four quadrants. The top-left and bottom-right quadrants are occupied by negative instances and the top-right and bottom-left quadrants by positive instances. The lines between the classes are the "ideal" boundary that separates the two classes.

Figure 6 exhibits the boundary distortion between the two left quadrants in Figure 5 under two different negative/positive training-data ratios. Figure 6(a) shows the SVM class boundary when the ratio is 10 : 1. Figure 6(b) shows the boundary when the ratio is 1000 : 1. The boundary in Figure 6(b) is much more distorted compared to the boundary in Figure 6(a), and hence causes more false negatives.

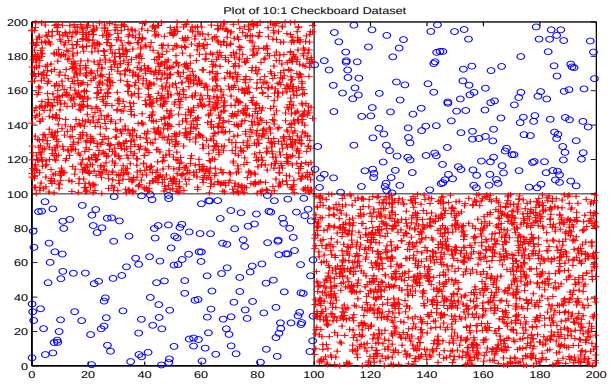
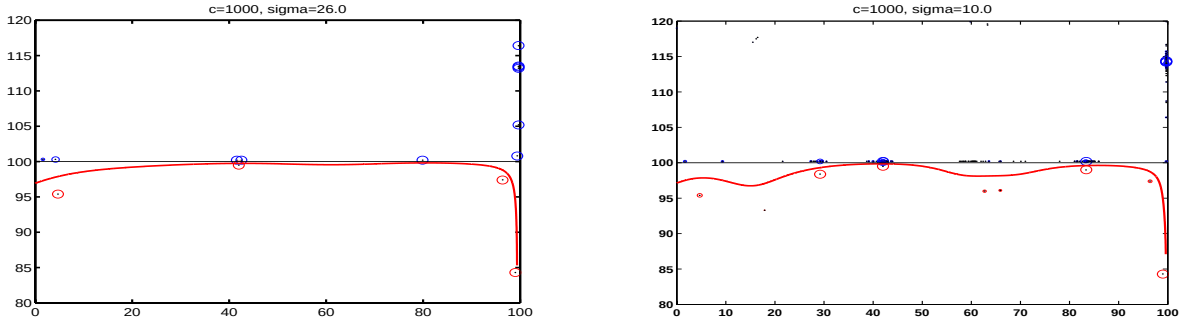


Figure 5. Checkerboard Experiment. Negative:Positive=10:1.



(a) 10:1 (b) 1000:1

Figure 6. Boundary Distortion.

2.2.3. Proposed Remedies

Before presenting our remedies, we provide two criteria for class-prediction evaluation.

1. *Tradeoff between specificity and sensitivity.*

   We define *sensitivity* of a learning algorithm as the ratio of the number of true positive (TP) predictions over the number of positive instances (TP+FN) in the test set, or

   $$Sensitivity = TP/(TP + FN). \tag{9}$$

   The *specificity* is defined as the ratio of the number of true negative (TN) predictions over the number of negative instances (TN + FP) in the test set, or

   $$Specificity = TN/(TN + FP). \tag{10}$$

2. *Overall accuracy.*

   Overall accuracy is the most general criteria of performance, defined as the ratio between the number of correctly identified predictions and the test set size, or

   $$OverallAccuracy = (TP + TN)/(TP + TN + FP + FN). \tag{11}$$

   Our design goal is to improve both sensitivity and overall accuracy.

In the following, we present four methods for achieving our design goal. The first two methods, *thresholding* and *penalty*, aim to tackle the problem of *unequal class importance*, making better tradeoff between specificity and sensitivity. The last two methods, *conformal mapping* and *sampling majority class*, help alleviate the drawback of *imbalanced training data*.

1. Making tradeoff between specificity and sensitivity.

   (a) *Thresholding method.*

   As discussed at the start of this section, a simple method for reducing false negatives is to change the decision threshold b in Equation 8. This shift trades specificity for sensitivity. The new decision function is

   $$f(x) = sign(\sum_{i=1}^{p} \alpha_i y_i K(x, x_i) + \hat{b}), \tag{12}$$

   where $\hat{b}$ is the new threshold after boundary movement. We use *thresholding* as the yardstick to measure how the other methods perform.

   (b) *Penalty method.*

   Veropoulos[10] uses a soft margin technique with SVMs for controlling the trade-off between false positives and false negatives. The basic idea is to introduce different penalty functions for positively and negatively labeled instances, in which a larger multiplier α_i is assigned to the class in which misclassification carries heavier cost. The Lagrangian formulation in Equation 3 is generalized with two penalty functions for false-positive and false-negative errors as follows:

   $$L_p = \frac{||w||^2}{2} + (C^+ \sum_{i|y_i=+1}^{p} \xi_i) + (C^- \sum_{i|y_i=-1}^{p} \xi_i) - \sum_{i=1}^{p} \alpha_i [y_i(w \cdot x_i + b) - 1 + \xi_i] - \sum_{i=1}^{p} \mu_i \xi_i. \tag{13}$$

   The dual formulation gives the same Lagrangian but with different α_i constrained by

   $$C^+ \geq \alpha_i \geq 0 \qquad if \quad y_i = +1, \ and \tag{14}$$

   $$C^- \geq \alpha_i \geq 0 \qquad if \quad y_i = -1. \tag{15}$$

   C^+ and C^- are the penalty for the positive and negative sides respectively. If C^+ is larger than C^-, fewer positive data would be misclassified as negative data; thus FN is reduced, and vice versa.

2. Tackling the problem of *imbalanced training data.*

   (a) *Conformal mapping.*

   The performance of SVMs largely depends on kernel selection. S. Amari and S. Wu[11] propose an information-geometrical method, which modifies a kernel to improve performance. The method increases separability of data in the feature space by enlarging the spatial resolution around the class boundary. Given a positive scalar function $c(x)$, a conformal transformation of kernel K can be written as

   $$\widetilde{K}(x, x') = c(x)c(x')K(x, x'). \tag{16}$$

   Generally, $c(x)$ has large values at support-vector locations and small values elsewhere. The function of $c(x)$ is selected in a data-dependent way by using the information of Support Vectors (SVs).[11] This method first trains SVMs with the primary kernel function K to obtain an initial set of SVs. It then modifies K according to Equation 16. SVMs are retrained with the modified kernel. The procedure iterates several times until prediction accuracy cannot be further improved.

   (b) *Sampling.*

   Kubat and Matwin[12, 13] address the imbalanced data problem by using a *one-sided sampling* method to selectively remove tuples from the majority class. Consequently, class distribution can be less imbalanced. One-side sampling may suffer from degraded prediction accuracy due to sampling variance. To remedy this problem, we incorporate *Bagging*[14] with one-sided sampling. Our sampling procedure is described as follows:

- Randomly sample a subset of the negative training data, and bag them together with the entire positive set in S_i.
- Train predictor $\bar{P}_i$ using S_i.
- Repeat the above two steps for N times. Get a set of predictors $\{\bar{P}_i, i = 1...N\}$.
- Perform voting to form the final predictor $\bar{P}$ for N bags,

$$\bar{P} = \frac{1}{N} \sum_{i=1}^{N} \bar{P}_i. \tag{17}$$

2.3. Calibration

The number of possible combinations of external factors such as camera capability and environment variables can be exponential. It is impossible to collect training data for every possible circumstance. For example, if the training data were collected in the daytime, the trained classifier based on the daytime data may not accurately detect events in the night, and vice versa. We remedy this training-data collection problem by calibrating the classifier.

To make the calibration scheme trackable, we assume that a camera is stationary except for limited zooming positions. Thus, the calibrator can pre-construct a number of background models. The classifier can then use the camera parameters obtained via self-calibration[15,16] to select a model to use. Once the background model is located, the calibrator compares the model to the live background and computes feature differences. The feature differences are fed back to the classifier to tune the event detection parameters. This calibration process will be discussed in detail in our future work.

3. EMPIRICAL STUDY

We have conducted experiments on detecting fire and smoke events to validate the effectiveness of our proposed learning methods. The experimental data are collected from movies that contain fire/smoke scenes, and from documentary video clips from fire stations. For SVMs, we use Gaussian kernel with $\gamma = 0.01$.

Type	$FN + FP$	FP	FN
Default SVMs	41.0%	22.2%	18.8%
Thresholding	37.0%	33.3%	3.7%
Penalty	35.0%	31.3%	3.7%
Conformal Mapping	30.3%	7.4%	22.6%
Sampling	26.9%	17.1%	9.8%
Combined Method	23.4%	19.3%	4.1%

Table 2. Comparison of Methods.

Table 2 summarizes the results in three columns, FN (false negative), FP (false positive), and FN+FP (overall error) for six different learning schemes. The first row in the table shows the prediction errors of SVMs. The second and third rows show the results of using the thresholding and penalty methods, respectively. As expected, the thresholding method sacrifices FP for improving FN. The total number of error rates remain about the same as that of SVMs.

For the penalty method, we first set both C^+ and C^- as the optimal C obtained from the SVM experiment. We then decreased the value of C^- to find the lowest FN point. Figure 7 shows the effect of reducing C^- from 100 to 0.7. The lowest FN occurs at $C^- = 10$. However, at the 3.7% FN level, the penalty method achieves only a 2% lower overall error rate than the thresholding method.

We first examined the best setting for the sampling method. Figure 8 shows the experimental result for different sampling rates with different numbers of bags. The best combination for our data is at 40% sampling rate with nine bags. The third group of the results presented in Table 2 show that both the conformal mapping

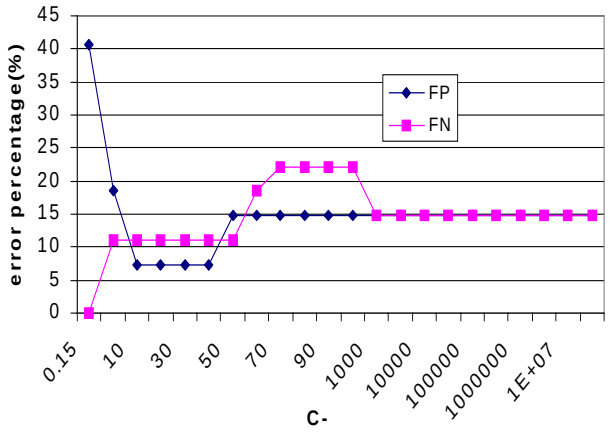


Figure 7. Choosing Different Penalty.

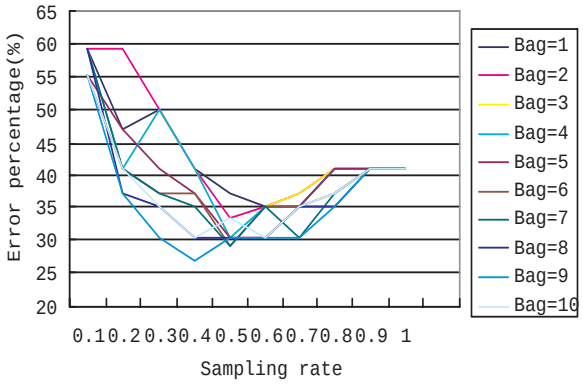


Figure 8. Sampling with Different Sampling Rates and Bag Numbers.

and the sampling method can reduce overall error significantly. The overall error rate drops from 41.0% to 30.3% for conformal mapping and to 26.9% for the sampling method. Although both methods are effective in reducing FP+FN, neither is able to effectively reduce FN. The table shows that the FN rates of these two methods are markedly higher than those of the thresholding and penalty methods.

This finding led us to experiment a combined scheme: We combined the penalty method with the sampling method to reduce both FN and FN+FP. The last row of Table 2 shows that the combined method is most effective in achieving our design goal: The penalty method focuses on reducing FN and the sampling method alleviates the skewed boundary problem caused by the imbalanced training data setting.

4. RELATED WORK

There has been a great deal of work in the area of video analysis. We discuss related work in *feature extraction*, and *video understanding*.

4.1. Feature Extraction

Features extracted for conducting video analysis can be categorized into *intra-frame features*, *inter-frame features*, and *multimodal features*.

- *Intra-frame Features.*
  Intra-frame features are perceptual cues extracted from individual video frames. Widely used intra-frame features include color, texture, shape, and layout information.[17] Among many, examples of commonly used features are pixel intensity,[18] region-based statistical characteristics of intensity,[19] intensity histograms,[20] texture co-occurrence matrix,[21] texture Gabor filter,[22] and edge change patterns.[23]
- *Inter-frame Features.*
  Inter-frame features such as temporal variance, object motion, and camera movement describe the temporal characteristics in a shot. Zhong[24] utilizes the variance of color histograms over all the frames in a shot to measure temporal variance. Chen[25] uses a 3D curve to represent motion tracks of objects. In the compressed video domain, DCT coefficients and motion vectors have been widely used to model motion patterns.[26–29]
- *Multimodal Features.*
  More recently, several researchers have extracted speech, closed-caption, and audio features to augment visual information. For example, the Informedia project[30] at CMU represents a pioneer effort in the use of speech recognition to produce a transcript for video retrieval. The Optical Character Recognition (OCR) technology has been used for extracting text in the absence of transcript and closed-caption data. Multimodal feature

extraction explores means to combine complementary information from different sources for analyzing videos more effectively.

For event detection, we may not be able to obtain information such as speech or audio, and hence we must solely rely on visual cues. As discussed in Section 2.1, our design goals are accuracy and speed, and our design challenge is to find features that can achieve a good balance between these two design objectives.

4.2. Video Understanding

The gap between low-level features and high-level semantics makes video understanding difficult. Many studies have been conducted for video understanding, including lecture/news summary,[31,32] motion tracking,[5,33] surveillance,[34] sports-event highlighting,[35–37] and movie-genre classification.[38,39] These studies can be divided into two categories of feature-to-concept inference: *intra-frame inference* for object recognition, and *inter-frame inference* for event identification.

- *Object Recognition.*
  Object recognition and semantical classification have been intensively studied for still images (e.g.,[40,41]). Pertinent to fire and smoke detection,[4,42–44] Walter's approach[42] creates a Gaussian-smoothed color histogram to locate suspect fire-colored pixels, and then examines temporal patterns of suspects to determine actually fire pixels. Wieser[43] ascertains the density of smoke by the loss of luminance in the background.
- *Event Detection.*
  Motion patterns of objects are essential for identifying events. A representative system is the vehicle surveillance system built.[5] The system first detects objects and tracks the moving patterns of the objects. From the moving speed and trajectory of the tracked objects, both object identity and behavior can be inferred. Another exemplary work is Datta's human violence detection.[34] The system first detects body limbs, and it analyzes motion trajectory and orientation about the limbs. Many studies reply on detecting repetitious motion segments for identifying a target event, e.g., Hamada's cooking event detection[33] and Zhong's baseball event detection.[36]

Built on the object recognition and event detection techniques discussed above, our work focuses on dealing with two statistical learning challenges: classes of unequal importance, and imbalanced in training data.

5. CONCLUSIONS AND FUTURE WORK

We have presented a general framework for hazardous event detection. Specifically in this paper, we describe our improved statistical learning methods for dealing with the challenges of unequal class importance and imbalanced in training data. There are a number of possible extensions of this work.

1. We plan to extend the hazardous event detection framework to cover other events such as suspicious activities and threatening behaviors.
2. We plan to tackle the problems of variable environments and camera capabilities. Since the amount of training clips is limited, we would like the classifier to generalize well in detecting hazardous events in unforeseen settings.

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